Interactive Virtual Museum of African Computing History

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ABSTRACT

This report outlines the development of an interactive virtual reality museum that immerses users in the evolution of computing within Africa — from early colonial influences to post-independence innovation and resistance technologies. By combining immersive narrative, user-driven exploration, and insights from decolonial computing, the project seeks to provide an entertaining and instructive experience in multiple themed exhibition halls. Through a study of the literature, an analysis of user requirements, and interface design, the research shows how VR may be used to maintain Africa's computer legacy, improve user engagement, and advance inclusive technological narratives.

KEYWORDS

Virtual Reality (VR), African Computing History, Decolonial Computing, Immersive Learning, Interactive Exhibitions.

1 INTRODUCTION

1.1 Project Background

African computing has a rich history, but traditional curricula frequently ignores it. A virtual reality (VR) museum showcasing significant computer breakthroughs in Africa is the goal of this project. The project uses immersive technology to create an interesting learning environment that encourages decolonial computing narratives and improves memory.

1.2 Project Description

This project involves the creation of a single-player virtual reality (VR) museum dedicated to African computing history. Users will be able to explore multiple themed exhibition halls, interact with 3D models of computing artifacts, access informational panels, and engage with puzzles and guided narratives. By leveraging VR technology, the museum provides an immersive and interactive learning environment that encourages engagement, promotes knowledge retention, and fosters appreciation of Africa's contributions to computing.

The need for this project arises from the broader trend of using immersive technologies in education, which have been shown to enhance experiential learning compared to traditional methods. Additionally, the museum challenges Eurocentric narratives in computing history by highlighting Africa's underrepresented technological achievements, creating both an educational and decolonial experience for users.

1.3 Objectives

The objectives of this project are:

- Create themed rooms that narrate distinct stories in African computing history.
- Integrate interactive elements such as puzzles and artifact handling.
- Implement narration, sound design, and dynamic lighting for immersion.
- Include an achievement system that rewards exploration and discovery.

1.4 Scope

This project will feature a single-player virtual reality museum with multiple exhibition halls, each focusing on a particular period or theme in African computing history. The VR application will include interactive 3D models, information panels, and a simple achievement system to enhance user engagement. The number of halls and their specific themes may evolve during development to better reflect historical content and user experience goals.

2 LITERATURE REVIEW

2.1 Virtual Reality in Education

Virtual Reality (VR) has been utilized extensively in the classroom to produce immersive experiences that let students interact with historical and cultural material in novel ways. Examples include virtual museum tours that allow students to visit famous museums like the Smithsonian or the Louvre from the comfort of their classrooms, as well as historical reconstructions that allow students to virtually experience life on a World War I battlefield or stroll through ancient Rome. Language learning becomes more realistic and engaging when students are able to experience the customs, traditions, and social norms of many countries using virtual reality. [1]

Compared to traditional teaching techniques, virtual reality has various advantages, such as enhanced engagement, since it engages students by letting them actively participate rather that passively receive information. Immersion and interactive learning

experiences improve comprehension of difficult topics and memory retention. Students can practice in a risk-free setting using VR's hands-on experience in disciplines like medicine, engineering, and emergency response. Furthermore, it overcomes geographic boundaries by allowing students to "travel" to many places, interact with diverse cultures, and obtain top-notch education regardless of their actual location. [1]

Studies show that virtual reality greatly boosts student engagement since it makes the, feel more linked to the material. Compared to regular classrooms, where memory recall rates are about 5%-10%, research has shown that VR can increase them to 75%-90%. According to study, students who utilized VR outperformed those who exclusively used video-based or textbook learning in terms of engagement and test scores [2], [3].

Cognitive benefits of VR include enhancing understanding and deepening cognitive processing by activating various senses [1]. VR promotes psychological involvement by making learning more exploratory and enjoyable [1]. Nevertheless some students might feel lightheaded or motion sick as a result of sensory imbalances in VR environments [3]. A further worry is the possibility of addiction, since some students might favor VR settings over in-person interactions [2] [3].

Challenges in using VR for education include high costs, as purchasing VR headgear, computers, and specialized software is expensive, making it inaccessible to many underfunded schools. Learning may be negatively affected by technological constraints, such as low resolution screens and lag in inexpensive VR systems. The distinction between visual cues and actual movement causes motion sickness, which is a common problem. Many teachers also lack the technological know-how to successfully incorporate VR into their curricula. The digital divide remains a major obstacle, as students from wealthier schools have greater access to VR, increasing educational inequalities [4], [1].

Best practices for VR-based learning include making sure that top-notch content is interactive, graphically clear, and created with clear learning objectives. Optimizing VR experiences can help minimize motion sickness by reducing lag and disorientation. For educators to successfully apply VR-based learning methodologies, sufficient teacher training is essential. To guarantee greater accessibility, institutions should look at affordable options like shared VR equipment or mobile-based VR experiences. Finally, using VR in conjunction with Artificial Intelligence (AI) can assist customize educational materials to meet the needs of specific students. [1]

These findings demonstrate that VR has a lot of educational potential, especially in the areas of engagement, retention, and experiential learning. However, there are still obstacles that stand in the way of its broad acceptance, i.e. cost and, accessibility, and technology.

2.2 Decolonial Computing and African Tech History

Decolonial computing is a critical approach that looks at computing as a colonial phenomenon and emphasizes its roots in colonial structures and knowledge systems. It examines who is creating and utilizing technology, where it is being used, and why, with an emphasis on the body-politics and geo-politics of knowledge production. By encouraging pluralistic, regional viewpoints, this viewpoint opposes the predominance of Eurocentric epistemologies. Technology education may promote inclusive innovation that prioritizes under-represented voices and opposes the expansion of colonial power systems by including decolonial viewpoints. [5]

For the most part, Eurocentric viewpoints have impacted the history of computing, obscuring the contributions of non-Western countries, such as Africa. African civilizations' indigenous knowledge systems and innovations are typically overlooked in favor of framing computing as a contemporary Western invention. The way that the history of technology focusses in Western accomplishments while ignoring local adaptations, innovations, and indigenous computer methods is another example of how African contributions are excluded. [5]

Significant contributions from Africa to computing and technology include mobile technologies like M-Pesa [6], which revolutionized financial transactions in the region, and telecommunications developments. M-Pesa is a mobile banking service that allows users to store and transfer money through their mobile phones. It was introduced in Kenya in 2007 by Safaricom, the country's largest telecommunications company, as a way to provide broader access to financial services. Safaricom, founded in 1997 and headquartered in Nairobi, offers mobile voice and data services, internet access, and enterprise solutions, and is partially owned by Vodafone Group and the Government of Kenya. Since 2020, M-Pesa has been a joint venture between Safaricom and Vodacom and now operates in several countries, including Kenya, Tanzania, Mozambique, Lesotho, Ghana, Egypt, and the Democratic Republic of Congo (DRC). M-Pesa has played a transformative role in advancing digital financial inclusion across the region [6]. Decolonizing participatory design has also been greatly aided by African academics and practitioners, who have made sure that technological advancements meet local needs. African researchers are also engaged in creating AI models that tackle regional issues, with focus on ethical AI development. [7]

By allowing viewers to experience history from a variety of viewpoints, especially those of historically under-represented groups, immersive storytelling in virtual reality has the potential to reinterpret historical narratives. Reconstructing historical histories that contradict existing Eurocentric narratives is made possible by it. Users can participate more sympathetically and interactively with historical events, traditional knowledge, and cultural heritage using virtual reality. By presenting indigenous perspectives in an interesting and approachable format, this acts as a potent instrument for decolonizing historical narratives. [7]

Incorporating decolonial viewpoints into technology education presents a number of difficulties. Integrating decolonial ideas into a curriculum is challenging due to institutional resistance, which is still a major obstacle in many academic institutions that prioritize Western epistemologies [5]. Furthermore, decolonial computing lacks adequate theoretical development and broad acceptance in mainstream education due to a lack of resources and representation

[5]. The ability of alternative knowledge systems to gain momentum is further limited by political and economic hurdles, as financing and governmental support are frequently biased towards Western technology paradigms [7].

Assuring accuracy and depiction in order to prevent historical narratives from being skewed by contemporary biases is one ethical consideration when presenting historical tales using virtual reality. To guarantee respectful and genuine storytelling, it is equally crucial to include the communities whose histories are being represented. Moreover, VR can be a potent emotional instrument, therefore it is important to be careful when portraying traumatic histories top prevent harm. [7]

By tackling these issues and moral dilemmas, decolonial computing can change how technology is learnt, created, and used, creating a more equal and inclusive online environment.

2.3 Past VR Projects

VR technology has proven its adaptability and efficiency by being used in a variety of fields, such as construction, education and fitness. Numerous initiatives have investigated how VR improves accessibility, user engagement, and collaboration while resolving usability and technological issues.

One such initiative, VR in Construction – Multi-User and Multi-Purpose, combines VR with Building Information Modelling (BIM) to enhance engineering, architecture, and construction collaboration. It accesses VR applications in eight real-world projects, showing multiuser VR sessions improve design reviews, foster communication, and assist in identifying constructability difficulties, prior to construction starting [8].

Similarly, the goal of Exploring the Potential of Virtual Reality in the Delivery of Personal Exercise is to employ VR to develop immersive training experiences that are suited to the needs of individuals. Physical therapists can prescribe training regimens and monitor users' progress using the project's integration of a VR application and a companion online app. The project emphasizes the value of user participation, safety, and accessibility in virtual workouts [9].

In the field of education, VR Architectural Visualization for Educational Institutions created a virtual reality program that lets students virtually tour a university campus. In order to help potential students virtually explore campus buildings, the team produced an immersive architectural visualization. According to the System Usability Scale (SUS), the VR tool was a good substitute for career counselling and campus tours [10].

Key takeaways from previous VR applications include greater comprehension of spatial ideas, increased engagement through interactive learning, and difficulties with motion sickness and device compatibility. Although VR makes learning more interesting, adoption hurdles like technological specifications and user adjustments need to be taken into account [10].

Typical technical problems in previous VR projects include data loss when transferring information between VR and external platforms like BIM software, hardware limitations that limit VR use to high-performance devices, motion sickness from latency and inconsistent motion tracking, and confusing user interfaces [8].

Projects have improved performance using methods like geometry instancing and occlusion culling to solve hardware and user interface issues, guaranteeing smoother frame rates. Frameworks such as Open VR have been used to increase cross platform compatibility. Furthermore, features like mini-maps, gesture-based controls, and drop-down menus improve accessibility and user navigation [8] [9].

Strategies to make VR applications more inclusive and engaging include multi-user capabilities to promote real-time collaboration, customizable surroundings with adjustable font size and brightness settings, and variable difficulty levels. These enhancements help to make VR more accessible and user-friendly [8] [9].

Past VR initiatives' best practices recommend utilizing multi-user features to improve interaction, streamlining performance for a more seamless experience, enhancing navigation with user friendly UI elements, and emphasizing accessibility with customizable options like voice navigation and seamless controls. By putting these best practices into practice, VR applications are certain to be inclusive and successful in a variety of fields [8].

2.4 Museums as a Domain for VR Applications

Virtual Reality (VR) is increasingly being applied in the museum domain to enhance visitor engagement, learning, and cultural participation. Studies show that VR can transform traditional museum exhibits into interactive and educational experiences, allowing visitors to explore collections in novel ways [11]. Bibliometric analyses highlight the growing use of immersive technologies, such as VR and Augmented Reality (AR), to enable storytelling and interactive exhibition design [12]. Research also indicates that museum visitors generally respond positively to VR, finding it improves accessibility and overall engagement with the exhibits [13]. These findings suggest that museums provide a unique context where VR can meaningfully combine education, heritage preservation, and visitor experience [14] [15].

3 USER REQUIREMENTS

3.1 Stakeholder Identification

Key stakeholders include:

- Students and Educators – For educational use.
- Historians and Researchers – For content accuracy.
- Local Institutions – For artifact curation and context.

3.2 Functional Requirements

- Users should be able to explore and interact with 3D artifacts.
- Puzzles should simulate historical computing activities.
- The system should track progress and achievements.

3.3 Non-Functional Requirements

3.3.1 Performance. To ensure a good VR experience, the application should operate smoothly. Interactions and loading times should be quick and not cause any discernible lag.

3.3.2 Security. User data must be managed safely in accordance with general data protection guidelines. The system must restrict access to sensitive information and, when necessary, preserve privacy.

3.3.3 Accessibility. The program should support a variety of user needs, such as preferred languages and fundamental visual and aural accessibility features. A broad spectrum of users, including those with little technological expertise, should be able to use it.

3.3.4 Reliability. The system needs to operate reliably in typical usage conditions. It should recover smoothly in the event of unforeseen faults without causing a substantial loss of data.

3.3.5 Usability. First-time users should find the application intuitive and easy to use. It should take very little direction to browse exhibitions or engage with important elements.

4 REQUIREMENT ANALYSIS

4.1 Data Requirements

- Exhibit Data: Historical details, artifact metadata, and interaction logic.
- User Data: Profiles, progress tracking, and achievement records.
- Puzzle Data: Challenge configurations, and solution validation mechanisms.
- Multimedia Data: 3D models, images, audio files, and textual descriptions.

4.2 Use Case Diagram

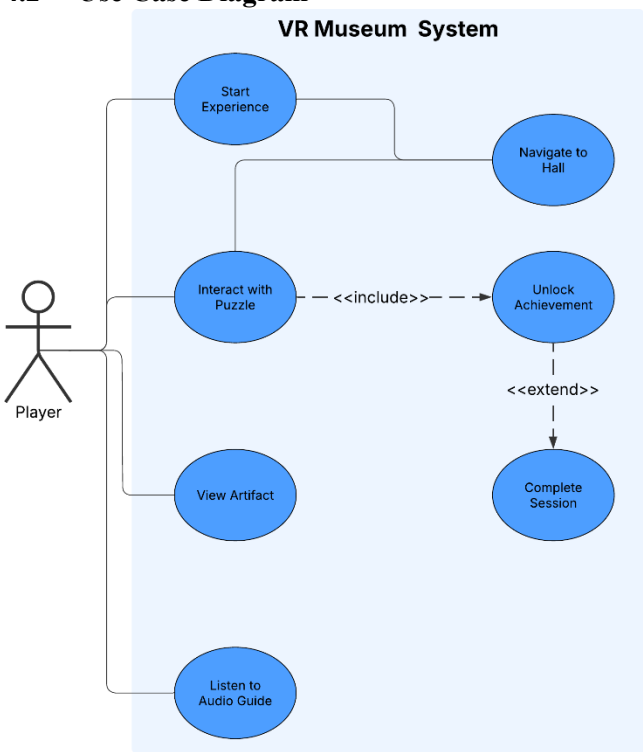

Figure 1: Use Case Diagram

4.3 Class Diagram

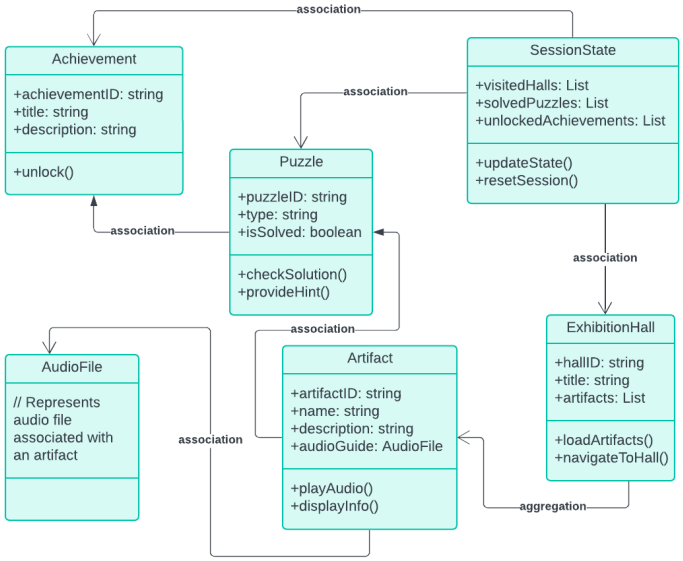

Figure 2: Class Diagram

5 USER INTERFACE SPECIFICATION

5.1 Navigation and Interaction

A controller-based movement system with guided paths, for accessibility, is used by users as they navigate through the virtual reality museum.

5.2 UI Design

- Information Panels: Styled to match the historical period (e.g., protest posters for Hall 2).
- Navigation: Markers styled to indicate cultural relevance (e.g., Ndebele-patterned markers).

6 RISK ANALYSIS

- Technical Risks: VR optimization challenges.
- Content Risk: Sensitivity of apartheid-related exhibitions.
- Adoption Risks: User engagement concerns.

7 DESIGN

7.1 Data/Class Design

The system follows an object-oriented approach as illustrated in the Class Diagram (see Figure 2). Unlike traditional persistent-user systems, this design eliminates permanent profiles and instead uses a SessionState class to manage temporary progress, such as solved

puzzles and unlocked achievements, within a single runtime session. Each Exhibit contains one or more Artifacts, some of which may include interactive Puzzles. Successful interaction may trigger Achievements, which are also tracked in-session. The ExhibitionHall class organizes artifacts spatially, and AudioFile objects are used for audio narration associated with specific artifacts.

7.2 Architectural Design

The system uses a modular, standalone VR architecture that runs locally on the user's headset, with no persistent data storage or login required. All progress is tracked temporarily during the session using a SessionState component.
It is structured into three layers:

- **Presentation Layer**: Handles 3D interface, user navigation, and input through Unity's VR environment.
- **Application Logic Layer**: Manages exhibit loading, puzzle interactions, achievements, and in-session state updates.
- **Content Layer**: Stores static resources like 3D models, audio narration, and exhibit metadata (e.g., JSON files).

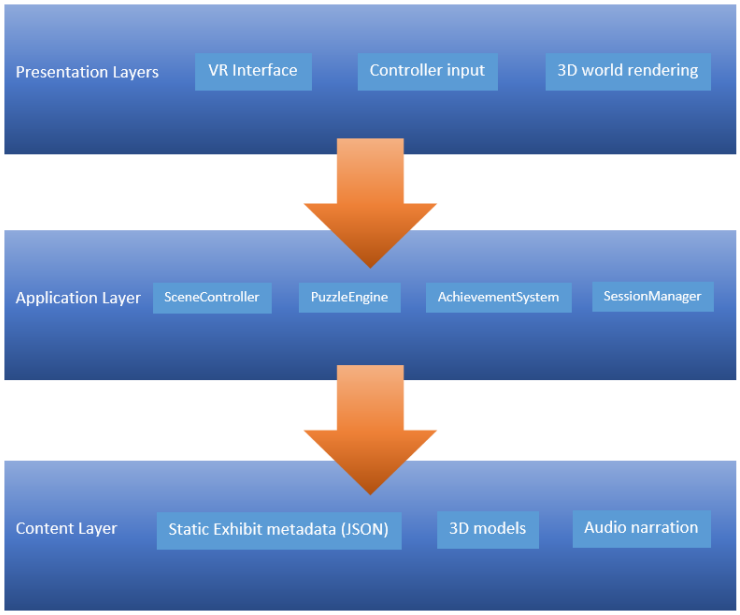

Figure 3: Layered Architecture Diagram

This layered design ensures a clear separation of concerns and simplifies maintenance while supporting a private, immersive, and user-friendly experience.

7.3 Interface Design

The interface is designed for intuitive use in VR, supporting natural navigation and interaction through controllers. Culturally themed cues (e.g., Ndebele patterns) guide the user through exhibit halls. Users interact with artifacts, view information panels, listen to audio guides, and engage with puzzles. Visual highlights indicate interactable objects, and achievements are displayed during the session as feedback.

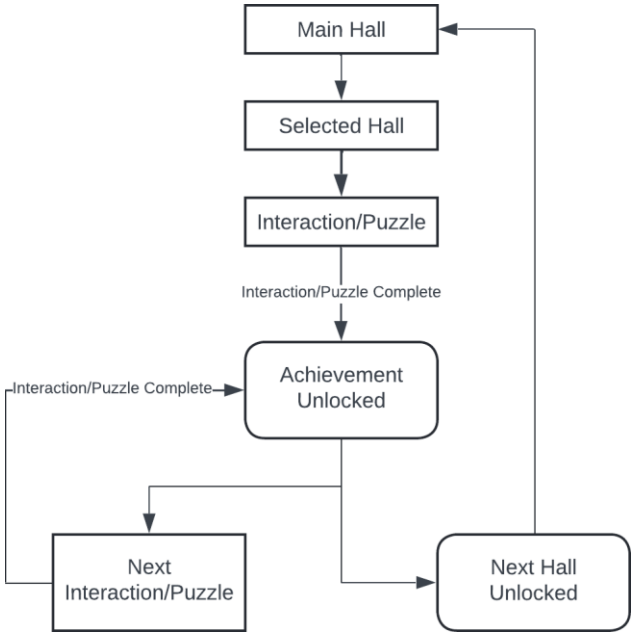

Figure 4: UI Flow Diagram

7.4 Component-level Design

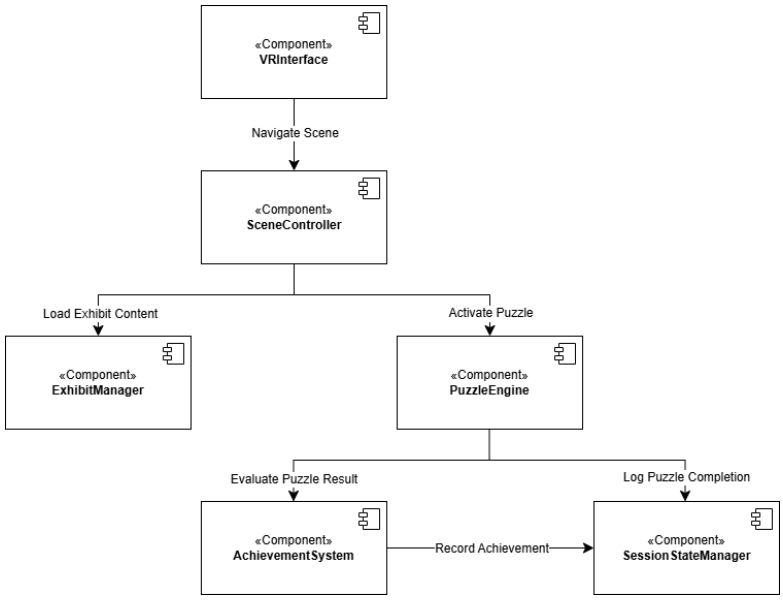

Figure 5: Component Diagram

The system is divided into modular components, each handling specific tasks:

- ExhibitManager loads and displays 3D artifacts.
- PuzzleEngine manages puzzle interaction and logic.
- AchievementSystem unlocks achievements based on puzzle completion.
- SessionState tracks progress within the current session.
- SceneController coordinates transitions and system flow.

These components communicate via event-driven mechanisms to keep interactions decoupled and maintainable.

8 PROTOTYPE

8.1 Hardware

The prototype is developed and tested using the following hardware:

- **VR Headset**: Oculus Quest 2
- PC Specifications:
  - Device name: E1-PC1
  - Processor: Intel(R) Core(TM) i7-6700K CPU @ 4.00GHz 4.00 GHz
  - Installed RAM: 32,0 GB
  - Device ID: 2C33EF5F-7F63-4F10-9D86-9FF5BC169A7A
  - Product ID: 00329-00000-00003-AA028
  - System type: 64-bit operating system, x64-based processor
- **Input Devices**: Oculus Touch Controllers

8.2 Software

The prototype software includes:

- **Development Engine**: Unity 2022 with XR Interaction Toolkit
- 3D Modelling Tools:
  - 3ds Max and Photoshop for artifact modelling and texturing.
  - Sketchfab for model downloads.
- **Audio**: Unity's built-in audio engine
- **Version Control**: Git and GitHub
- **Platform**: Oculus Quest (standalone build and PC-connected via Oculus Link for testing)

9 IMPLEMENTATION

9.1 Software and Hardware

The required hardware and software were outlined in the Prototype section (see section 8). These include the Oculus Quest 2 headset, development PC, Unity 2022 with XR interaction toolkit, C#, 3D modelling tools (3ds Max, Photoshop, Sketchfab), and GitHub for version control.

For implementation, these resources have been actively used to build and test interactive features. At this stage, no new additional hardware or software beyond those listed in the prototype has been required. However, further optimizations may require additional Unity packages or third-party assets during refinement.

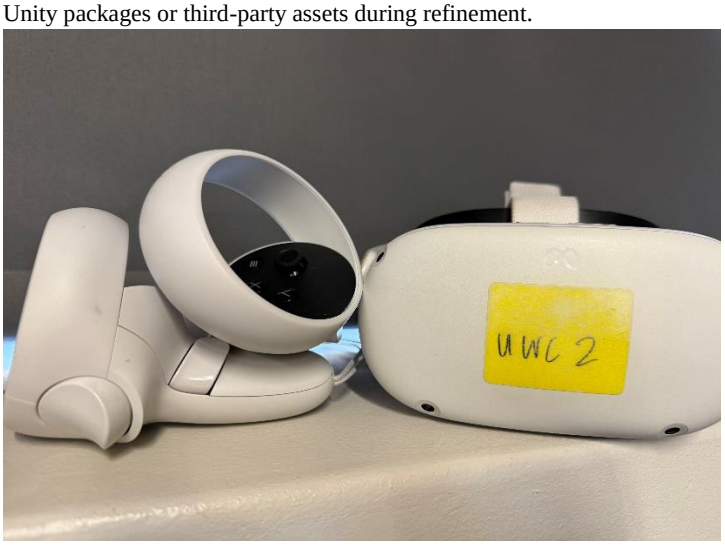

Figure 6: Oculus Quest 2 Headsets

9.2 Functions/Methods/Classes

Implementation followed an incremental approach, with features being developed and tested hall by hall. The system is not yet fully complete, but several functional modules have been implemented:

- Navigation Module: Handles movement and teleportation.
- Puzzle Interaction Module: Handles puzzle logic, activation, and validation.
- Virtual Keyboard Module: Handles user text input and puzzle solving.

Still to be implemented:

- Achievement System Module: To handle player's achievements.
- Session State Module: To track player's temporary progress.
- Scene Controller Module: To manages transitions of scenes.

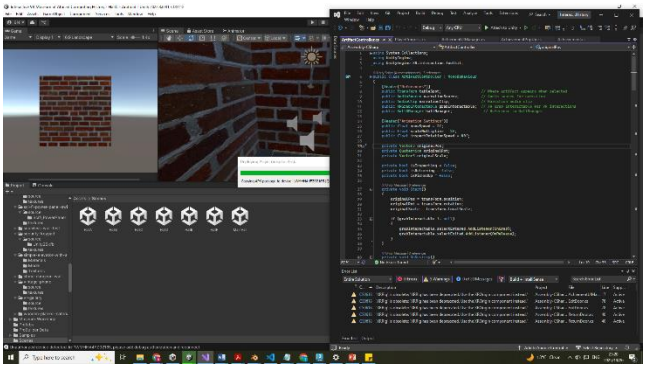

Figure 7: Unity

9.3 Testing

Since implementation is ongoing, only preliminary testing has been conducted to verify basic functionality:

- Navigation: Ensured teleportation and guided path work, though further refinement is needed to correct rotation alignment issues.
- Interactions: Confirmed that objects can be interacted with; polishing for a more interactive experience is planned.
- Puzzles: The Caesar puzzle functions correctly, but additional puzzles are still under development. Achievement system is also still getting set up.
- Audio: Initial narration works, though an issue with placement of the narration continuing across halls has been identified and is being addressed.
- UI: UI is activated along with audio but the placement of the text is still being improved.
- Performance: Early performance checks on Oculus Quest 2 indicate stable frame rates (>70 FPS) but optimisation will continue.

10 PROTOTYPE

Testing was conducted to ensure that the VR museum environment functions correctly, meets user requirements, and provides a smooth and educational user experience. Testing focused on interaction mechanics, scene navigation, performance on the Oculus Quest 2, and narrative delivery. Both verification ("are we building the product right?") and validation ("are we building the right product?") principles were applied.

10.1 Test Plan/Strategy

A structured testing strategy was followed:

Table 1: Testing Strategy Overview

Level	Purpose	Status
Unit Testing	Scripts (drawers, buttons, waypoints, etc.)	Complete
Integration Testing	Full hall interaction flow	Complete
Validation Testing	Meets functional requirements	Complete
System Testing	Overall VR experience stability	Complete
UX Testing	Comfort, accessibility, clarity	Complete

Testing was conducted in accordance with Principle 1, ensuring that all tests were mapped to the requirements, and Principle 4, which advocates starting with small-scale tests and progressively scaling up.

10.2 Test Design

Tests were designed around:

- Functional Requirements
- Interaction Requirements
- Navigation Requirements
- Educational Content Delivery
- VR comfort & safety

Test data included:

- User movement patterns through halls
- Object interactions (grabbing, pressing buttons, animations)
- Audio-trigger validation
- Achievement unlock states

10.3 Test Cases

Table 2: Core Interaction & Navigation Test Cases

TC #	Test Case Description	Expected Result	Status
TC-01	Teleportation activates only on valid waypoints	Player moves to target location	Passed
TC-02	Drawer opens when interacted with	Smooth open animation	Passed
TC-03	Artifact enlarges and narration starts when selected	Object displays on table + correct audio plays	Minor issues; audio problems
TC-04	Switch activates hall lights	Room illumination toggles correctly	Passed
TC-05	Book trigger interaction works	Hidden room unlocks	Passed
TC-06	Surveillance cameras detect the player	Trigger warning & consequence	Passed
TC-07	Disable surveillance system via lever puzzle	Cameras fully deactivate	Minor issues, lever not functional
TC-08	Elevator doors open/close correctly using buttons	Smooth synced animations	Passed
TC-09	Waypoints hidden when interacting with artifacts	Immersive interaction restored after	Minor issues with some waypoints
TC-10	Achievement unlock when hall objectives completed	Achievement stored	Passed

Table 3: Performance & Comfort Testing

Test	Criteria	Result
Frame Rate Stability	≥ 72 FPS on Quest 2	Passed
VR Safety Boundaries	No clipping into walls; teleport prevents discomfort	Passed
Motion Sickness Risk	No artificial locomotion	Passed

Table 4: Educational Delivery Testing

Requirement	Validation Method	Result
Historical accuracy	Peer review of content	Passed
Player engagement	User feedback surveys	Passed
Audio clarity	Testing narration triggers	Passed

10.5 Test Report

Testing confirmed that the system meets:

- All functional requirements
- Core VR usability expectations
- Educational goals
- Smooth performance on target device

Table 5: Issues Encountered & Resolved

Issue ID	Validation Method	Result
I-01	Teleportation points	Resolved
I-02	Drawer animation delays	Resolved
I-03	Artifact narration overlap	Resolved
	Puzzle unlocking hidden room	Resolved
	Surveillance detection	Resolved
I-03	Lever disables cameras	Minor issue, not yet functional
I-04	Waypoint visibility	Resolved
I-05	Elevator doors	Resolved

Test Summary (Final Acceptance):

- Total Test Cases: 10
- Passed Test Cases: 7
- Minor Issues: 2 (audio issues, camera deactivation and waypoint issues)
- Issues Identified: 8 (all addressed or minor)
- Performance Criteria: Met, with minor frame dips under heavy interactions
- Requirement Validation: All validated; historical accuracy, engagement, and audio confirmed

11 CONCLUSION

This project is a major step in changing the way that African computing history is taught and learnt. In addition to improving educational engagement, the initiative promotes a decolonial approach to technological storytelling by utilizing immersive virtual reality technologies. Users can better comprehend Africa's underappreciated computing history, especially during crucial periods like apartheid resistance and post-apartheid innovation, through themed exhibition halls, interactive features, and guided narratives. Technical concerns, accessibility concerns, and historical sensitivities provide difficulties, but the project's design prioritizes inclusion, usability, and educational value. The ultimate goal of this museum is to preserve and celebrate Africa's rich computing heritage while promoting a more inclusive technological conversation.

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